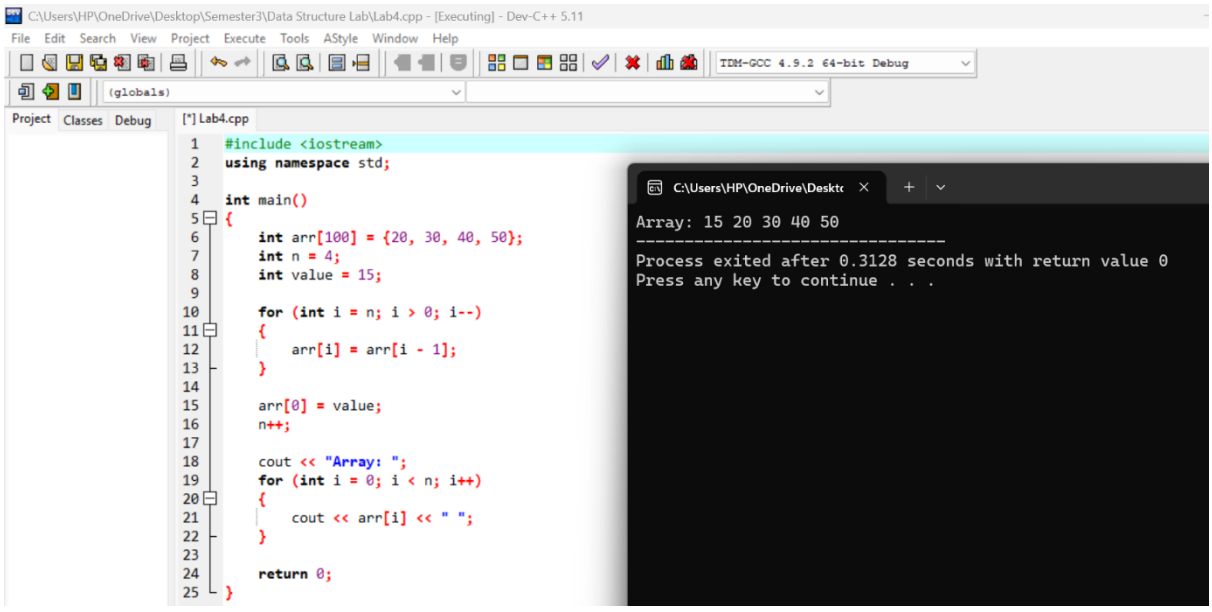
Name : Adeeba Bibi

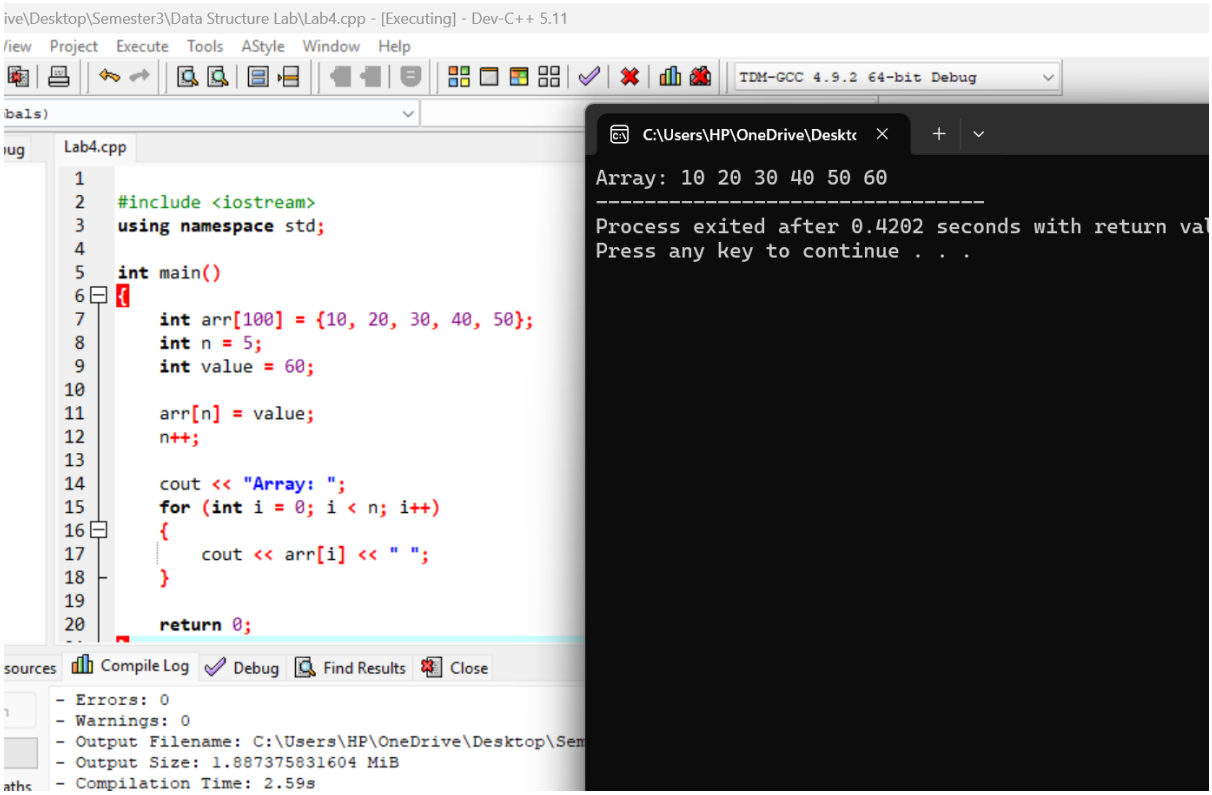
Sap Id : 73716

Lab : 04
Data Structure and Alogrithm
Array Operations

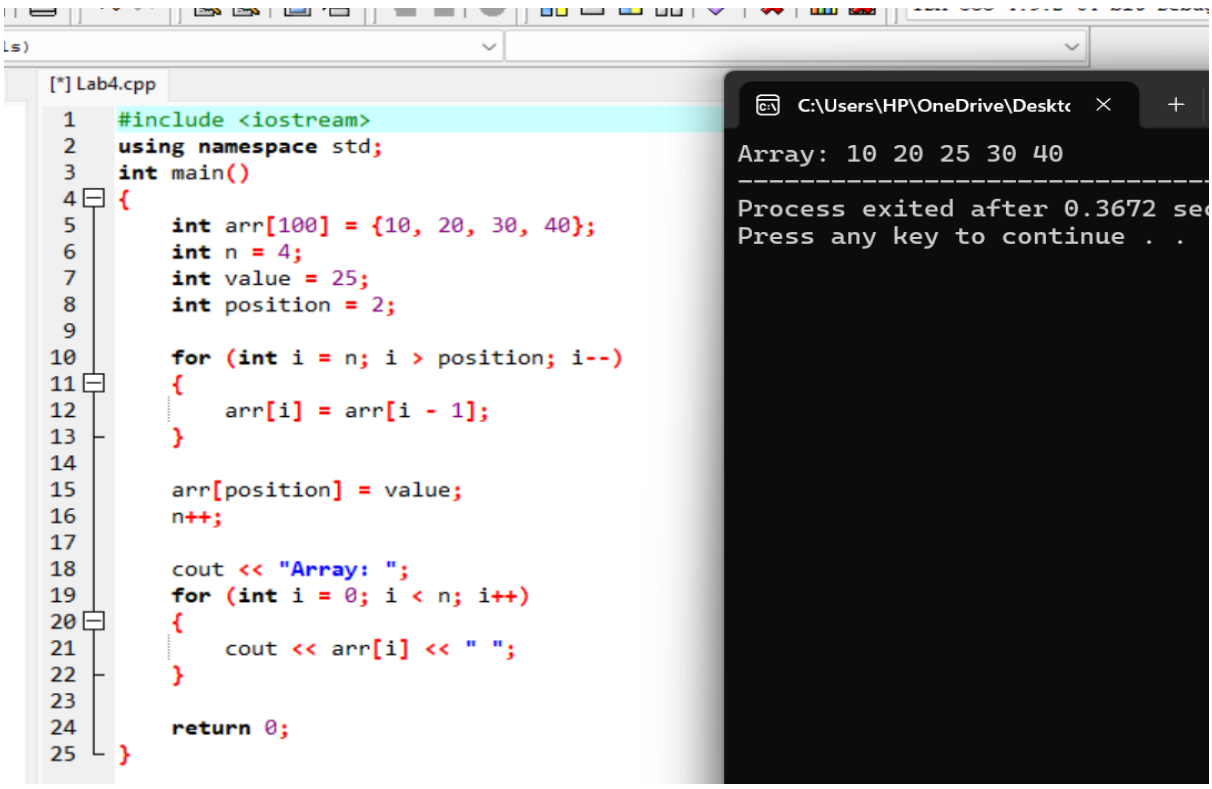
Task1.Insert 15 at the beginning of the array: 20, 30, 40, 50



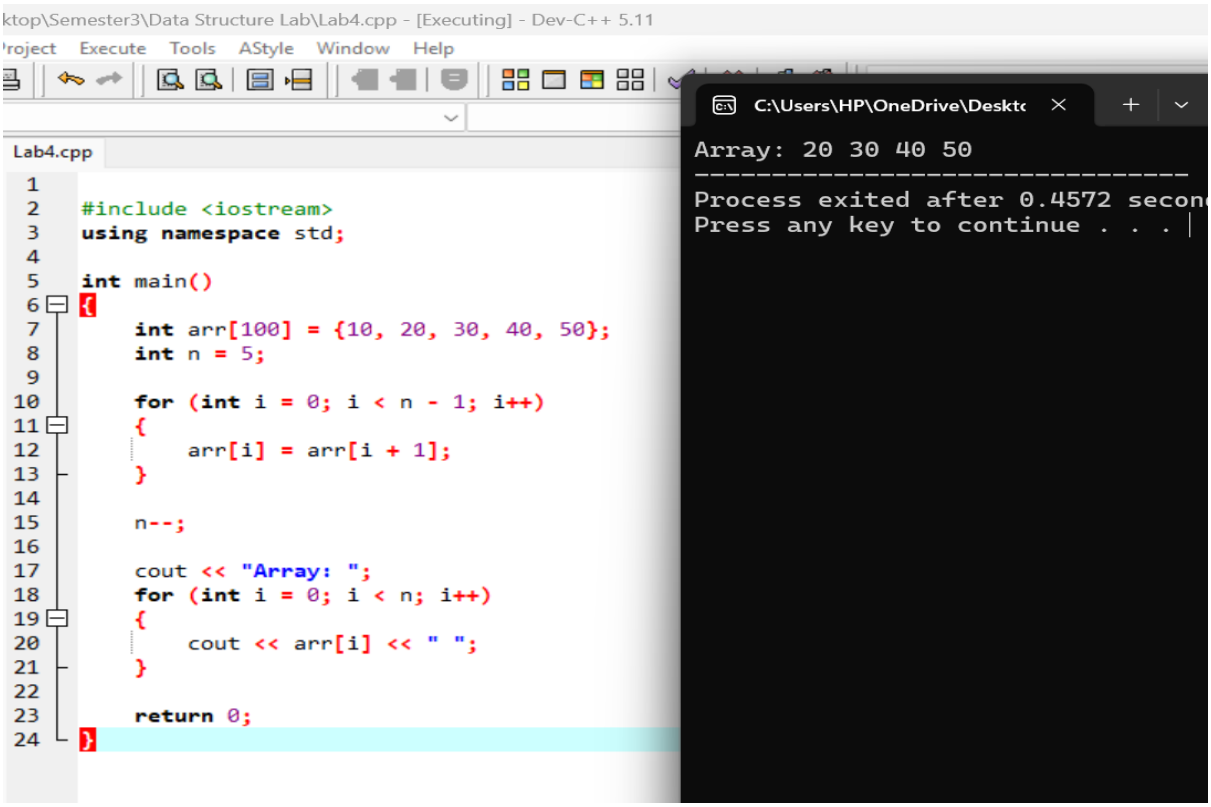
Task 2: Insert 60 at the end
Given array: 10, 20, 30, 40, 50



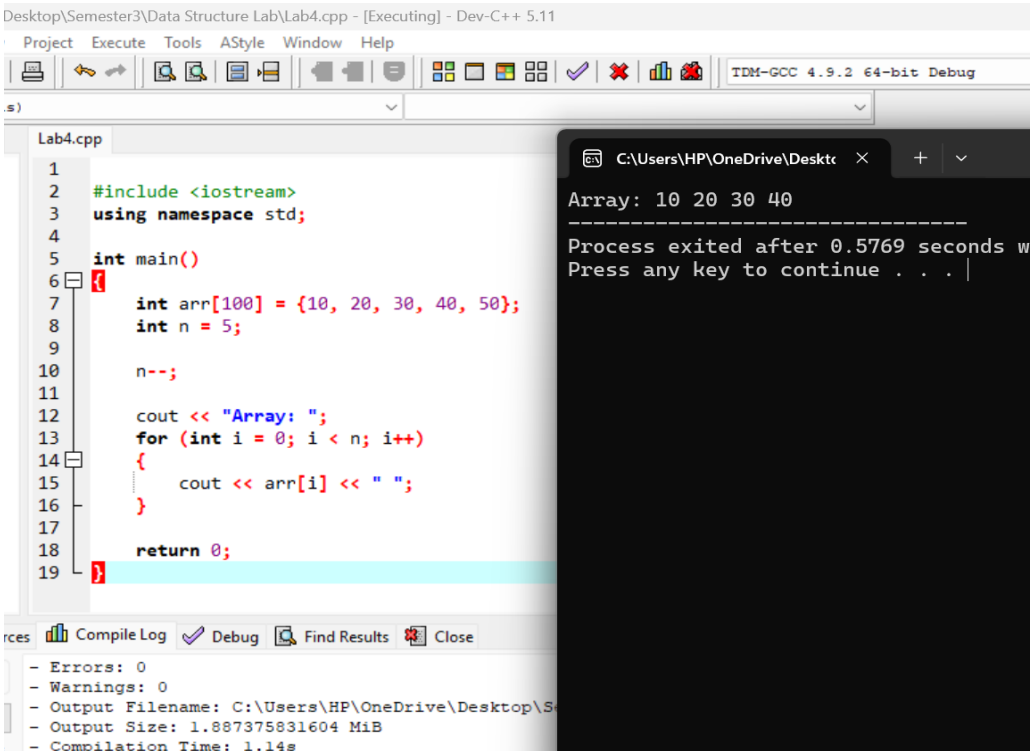
Task 3: Insert 25 at index 2
Given array: 10, 20, 30, 40



Task 4: Delete the first element
Given array: 10, 20, 30, 40, 50

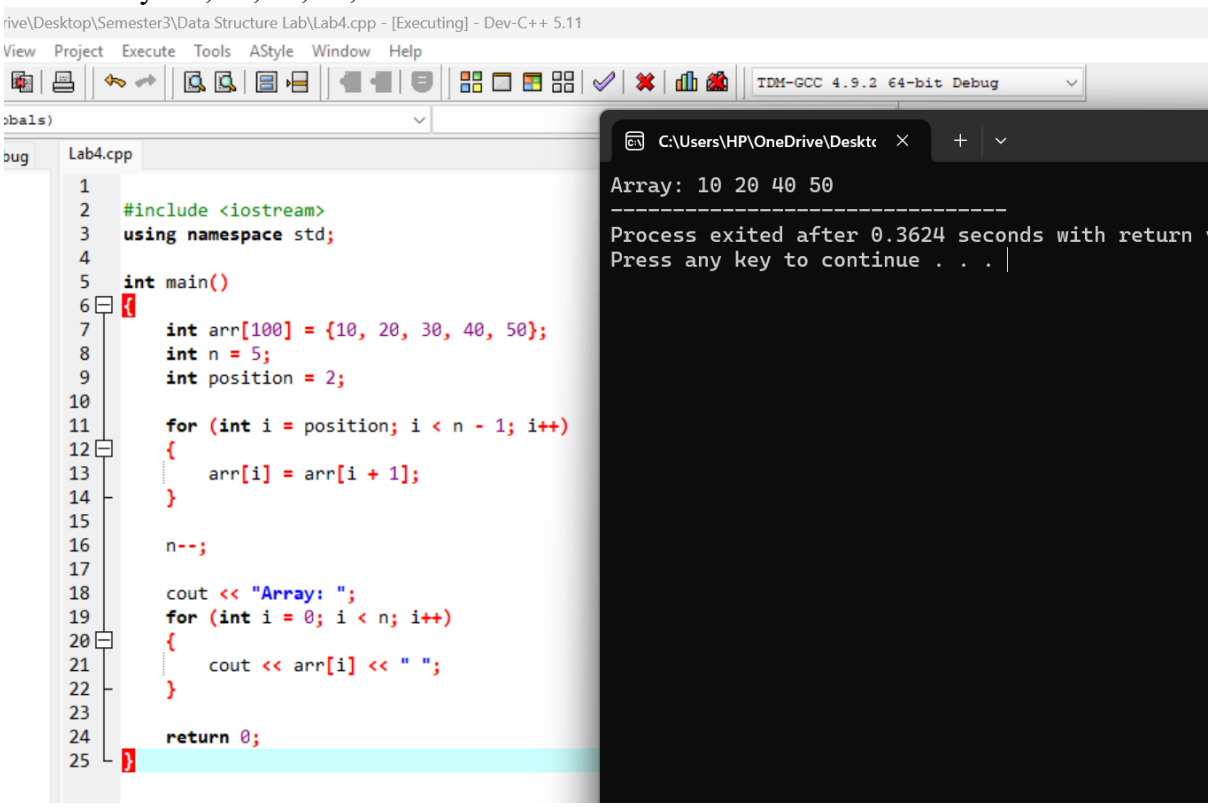


Task 5: Delete the last element
Given array: 10, 20, 30, 40, 50



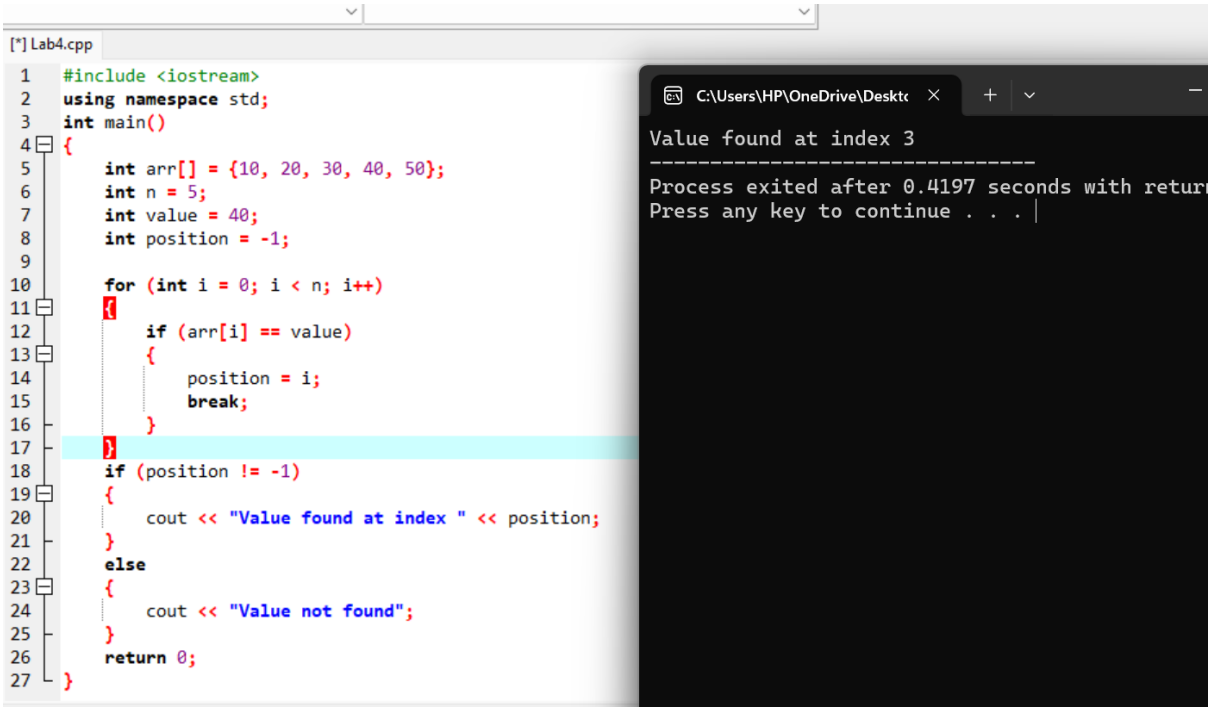
Task 6: Delete the element at index 2

Given array: 10, 20, 30, 40, 50

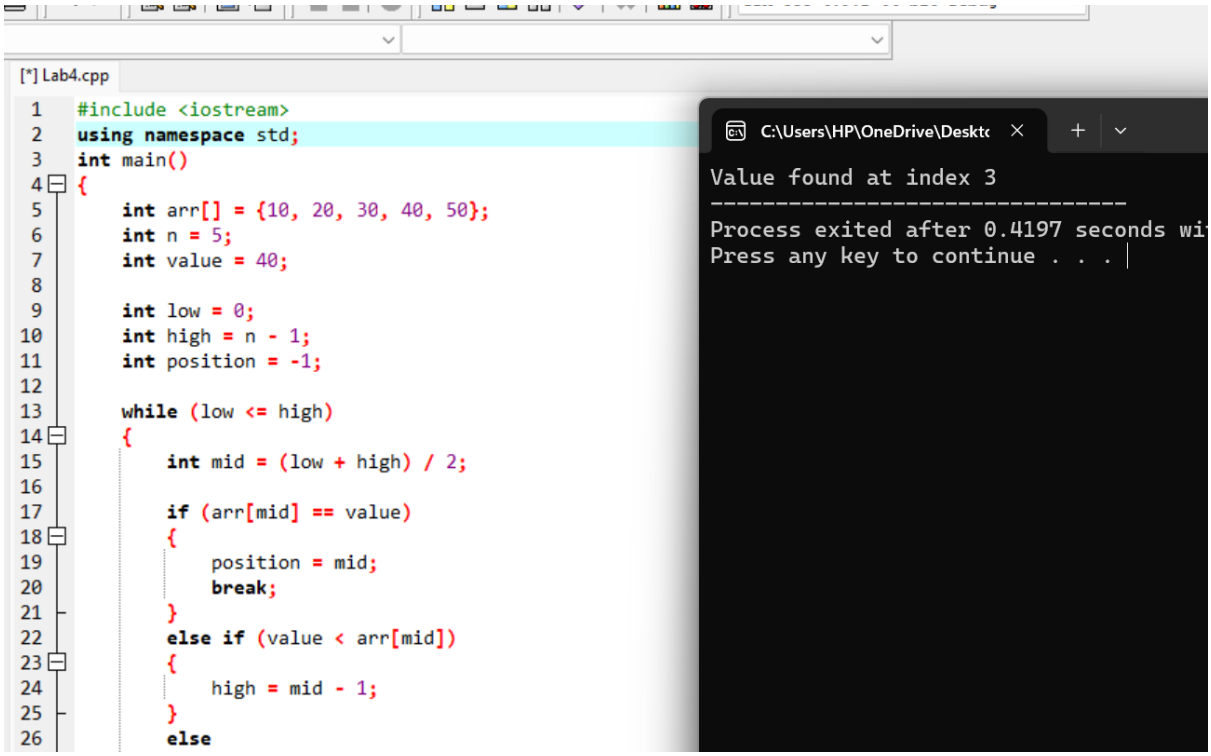


Task 7: Find 40 using Linear Search

Given array: 10, 20, 30, 40, 50



Task 8: Find 40 using Binary Search
Given sorted array: 10, 20, 30, 40, 50




```cpp
#include <iostream>
using namespace std;
int main()
{
    int arr[] = {10, 20, 30, 40, 50};
    int n = 5;
    int value = 40;

    int low = 0;
    int high = n - 1;
    int position = -1;

    while (low <= high)
    {
        int mid = (low + high) / 2;

        if (arr[mid] == value)
        {
            position = mid;
            break;
        }
        else if (value < arr[mid])
        {
            high = mid - 1;
        }
        else
        {
            low = mid + 1;
        }
    }

    if (position != -1)
    {
        cout << "Value found at index " << position;
    }
    else
    {
        cout << "Value not found";
    }

    return 0;
}
```